XI. Point Estimation: Introduction and Method of Moments (GSE 518)

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1. Point Estimation: Introduction

1.1. Setting

- Assume we have an iid sample $(X_1, \ldots, X_n)$.
- Assume X_i follows a specific pdf/pmf $X_i \sim f(x|\theta_0)$, where θ_0 is a parameter vector, i.e., the parameters of the distribution $f(\cdot)$.
- We assume that $f(\cdot|\theta_0)$ is *known* and $\theta_0 \in \Theta$ (parameter space) is *unknown*.

Example 1: Assume $X \sim N(\mu_0, \sigma_0^2)$. Here, $\theta_0 = \{\mu_0, \sigma_0^2\}$ and $\Theta = \{\mu \in \mathbb{R}, \sigma^2 > 0\}$.

Example 2: Assume $X \sim Gamma(\alpha_0, \beta_0)$. Here: $\theta_0 = \{\alpha_0, \beta_0\}$ and $\Theta = \{\alpha, \beta > 0\}$.

Goal: Estimate (or "guess") the value of θ_0 based on our sample $(X_1, \ldots, X_n)$: *Point Estimation*

We will focus on two main topics:

1. Methods to Estimate θ_0 (Notes XI and XII)
2. Methods to Evaluate Estimators (Notes XIII)

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

1.2. Definition of Point Estimator

Definition: Let $(X_1, \ldots, X_n)$ be a sample with joint pmf/pdf $f(x_1, \ldots, x_n|\theta_0)$ with $\theta_0 \in \Theta$. A function of the sample $\hat{\theta}(X_1, \ldots, X_n)$ that will be used to make inferences about θ_0 is called a *point estimator* of θ_0, and the value of the estimator for a particular sample $X_1 = x_1, \ldots, X_n = x_n$, or $\hat{\theta}(x_1, \ldots, x_n)$, is called the *point estimate* of θ_0.

- Notes on "Point Estimator":

  - Example: Assume $X \sim N(\mu_0, \sigma_0^2)$, and let $\theta_0 = E(x) = \mu_0$. Further assume that our sample is given by $(X_1 = 5, X_2 = 3, X_3 = 2, X_4 = 1, X_5 = 4)$ and we use the sample mean $\overline{X}$ to estimate μ_0.

    * The *Point Estimator* is: $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$
    * The *Point Estimate* is: $\overline{X} = \frac{1}{5}\sum_{i=1}^{5} x_i = \frac{15}{5} = 3$

  - Key Note: The parameter is θ_0 is *non-random* or fixed, but it is unknown. The estimator $\hat{\theta} = \hat{\theta}(X_1, \ldots, X_n)$ *is random* because it is a statistic, i.e., it is a function of the random sample $(X_1, \ldots, X_n)$.

    * Hence,$\hat{\theta}$ is going to have a distribution and all features associated with it, e.g., $f_{\hat{\theta}}(\hat{\theta})$, $E(\hat{\theta})$, $Var(\hat{\theta})$, $Pr(\hat{\theta} \leq r)$, etc.
    * Why is $\hat{\theta}$ random? (see lecture for intuition.)

  - Note on notation: Let $X \sim f(x|\theta_0)$ with $\theta_0 \in \Theta$. Then:

    * θ_0: Tue value of the parameter, $\theta_0 \in \Theta$
    * θ: *Any* element of Θ
    * $\hat{\theta}$: Estimator of θ_0 (it's random!)

    (See Examples from section 1.1.)

1.3. Finding Estimators

How can we use our random sample $(X_1, \ldots, X_n)$ to find estimators for θ_0?

Next, we analyze two of the most common approaches to: *Method of Moments* (rest of this set of notes) and *Maximum Likelihood* (Notes XII).

2. Method of Moments (MOM)

2.1. Motivation

To motivate the MOM approach, let $X \sim f(x|\theta_0)$ and let θ_0 be a scalar. Hence, our goal is to estimate or "guess" the value of θ_0. The logic of the MOM is the following:

- First, we know that $E(X)$ is a function of the parameter θ_0 (i.e., of the object we are after.)
  - Example: Let $X \sim \text{Beta}(\alpha_0, 1)$. Then, we know $E(X)$ is given by:

$$E(X) = \frac{\alpha_0}{\alpha_0 + 1}$$

- Second, we have a "good" estimator of $E(X)$, the sample average $\overline{X}$. Say,
  (i) $E(\overline{X}) = E(X)$, i.e., the distribution of $\overline{X}$ is centered at $E(X)$.
  (ii) As the sample size increases (i.e., as $n \to \infty$), $\overline{X}$ "gets closer" to $E(X)$ (later).
- **MOM's Idea:** Therefore, given the two points above, the main idea of the *MOM estimator*, $\hat{\theta}_{MOM}$, is to set set $\overline{X} = E(X)$ and solve for θ_0 (what we are after).

  Intuition: If $\overline{X}$ is "close to" $E(X)$ as $n \to \infty$, we expect (hope) that our MOM estimator $\hat{\theta}_{MOM}$ is also "close to" θ_0.

– Example (Continuation): In the example above, to get the MOM estimator of α_0 we set $\overline{X} = E(X)$:

$$\overline{X} = \frac{\alpha}{\alpha + 1}$$

and solve for α as a function of $\overline{X}$.

$$\hat{\alpha}_{MOM} = \frac{\overline{X}}{1 - \overline{X}}$$

Intuition: If $\overline{X}$ is "close to" $E(X)$ as $n \to \infty$, we expect (hope) that our MOM estimator $\hat{\alpha}_{MOM}$ is also "close to" α_0.

2.2. Method of Moments Estimator: Definition

Definition: Let $(X_1, \ldots, X_n)$ be an iid sample from $f(x|\theta_0)$, with $\theta_0 = \{\theta_1, \ldots, \theta_k\}$ (k parameters). Let

$m'_j = \frac{1}{n} \sum_{i=1}^{n} X_i^j$ for $j = 1, 2, \ldots, k$ (first k *sample* moments)

$M'_j = E[X^j]$ for $j = 1, \ldots, k$ (first k *population* moments)

Then the MOM estimator $\hat{\theta}_{MOM}$ of θ_0 is the solution to the system of k equations

$$m'_j = M'_j \quad \text{for } j = 1, \ldots, k.$$

Notes on "MOM Estimator":

- In words: Set the sample moments equal to the population moments (which are functions of our parameters) and solve for the θ's in terms of $(X_1, \ldots, X_n)$
- How many moments do we use? In general, we set

  Number of moments used = number of parameters that we estimate

so that: Number of equations = number of unknowns (to get a solution)

Hence:

- To estimate 1 parameter we set:

$$\overline{X} = E(X)$$

- To estimate 2 parameters we set:

$$\overline{X} = E(X)$$

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 = E(X^2)$$

- Example: Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} N(\mu_0, \sigma_0^2)$. Find $\hat{\theta}_{MOM}$.

  (See solution from lecture.)

- Notation: Hereafter, let

$$\widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

  Note that this is different from the sample variance S^2 we analyzed in the previous notes, which was defined as:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

- Key Note: We typically use *centered moments* (at mean) instead of uncentered moments (the ones used above). Specifically, if we have two parameters to estimate, we would set

$$\overline{X} = E(X) \quad \text{and} \quad \widetilde{S}^2 = Var(X) = E[(X - E(X))^2]$$

instead of

$$\overline{X} = E(X) \quad \text{and} \quad \frac{1}{n}\sum_{i=1}^{n} X_i^2 = E(X^2)$$

- The reason is that the centered moments are usually available.
- Careful: Do not mix them up! Say, **do not set:**

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 = Var(X) \quad \text{or} \quad \widetilde{S}^2 = E(X^2) \qquad \text{(do NOT use these equations!)}$$

- Example: Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} Binomial(n_0, p_0)$. Find $\widehat{\theta}_{MOM}$.

  (See solution from lecture.)

- Drawbacks of MOM Estimator:
  - It may not be efficient in some applications (later).
  - It may give estimates that are outside of parameter space Θ in some applications.
    * For example, in the binomial example above we know that $p \in [0, 1]$, but if $\widetilde{S}^2 > \overline{X}$ then we may have $\widehat{p}_{MOM} < 0$.

- The *Generalized Method of Moments* or *GMM*, which is widely used in econometrics and other fields, is an extension of the MOM idea.
  - GMM considers the case when the number of moments (or equations) is *greater than (or equal to)* the number of parameters (or unknowns).

    In this case we say the model is *over-identified*.
  - Intuitively, the GMM estimator $\widehat{\theta}_{GMM}$ equals the value of the $kx1$ vector θ that makes the square of the distance between the sample and population moments as close to zero as possible. In other words:

    * It makes the sample moments as close to the population moments as possible.
    * It minimizes the square of the distance between the sample and population moments

- Many estimators in econometrics can be seen as GMM or MOM estimators. For example:
    * OLS / Linear Regression
    * Logit and Probit
    * Poisson Regression
    * Quantile regression, etc.

- The "GMM form/representation" of estimators is very useful to derive some of their asymptotic properties, such as deriving their asymptotic distribution (say, approximating their distribution as $n \to \infty$) (later).